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REVIEW PAPER

A Meta-Analysis of Applied Behavior Analysis-Based Interventions for Individuals with Autism Spectrum Disorders (ASD) in Turkey

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Abstract

Interventions based on the principles of applied behavior analysis (ABA) have been determined to be evidenced-based practices and are widely used with individuals with autism spectrum disorders (ASD) across the world. Originally developed in the USA, implementation of these interventions has become widespread in the last decade in Turkey. Given the significance of culture in guiding ABA practices, this meta-analysis investigated the prevalence and the magnitude effects of ABA-based interventions and whether specific participant and intervention characteristics moderated such effects in Turkey. Seventy-one ABA-based intervention studies were assessed based on What Works Clearinghouse (WWC, Kratochwill et al., *Remedial and Special Education*, 34(1), 26–38, 2013) design standards (DSs) including (a) systematic manipulation of independent variables, (b) interrater reliability, (c) three demonstrations of the intervention effect, and (d) adequate number of data points collected for each condition. Forty-three studies that met WWC (Kratochwill et al., *Remedial and Special Education*, 34(1), 26–38, 2013) DSs were analyzed and calculated with baseline-corrected tau (Tarlow, *Behavior Modification, 41*(4), 427–467, 2017). Furthermore, performance-criteria-based effect size values (PCES; Aydin and Tanious, *Journal of Applied Behavior Analysis*, 2021) were calculated for 27 studies that had mastery criteria. While the overall tau appeared to be large 0.71 ($p = 0.072$, se = 0.254), the overall effect size for PCES revealed a small effect of 0.73. The comparisons of all sub-categories' effect sizes of moderator variables were not statistically significant based on Kruskal Wallis or Mann Whitney U tests. Additionally, findings revealed that studies were predominantly implemented by highly trained researchers in segregated one-on-one settings, and did not include adults with ASD nor challenging behavior. Overall, there has been significant growth in ABA-based interventions in Turkey that show promise in improving the lives of individuals with ASD. However, further research and cultural considerations are essential for a comprehensive understanding of their impact in the Turkish context. Findings from two different effect size analyses and recommendations for future studies are discussed.

Keywords Applied behavior analysis · Individuals with ASD · Meta-analysis · Turkey

Introduction

Autism spectrum disorder (ASD), a neurodevelopmental condition that appears in early childhood, is manifested by persistent challenges in social communication and interaction with restrictive and repetitive behaviors along with a narrow area of interest (American Psychiatric Association [APA] 2013). Various behavior problems such as self-injurious behavior (Matson & LoVullo, 2008; Smith & Matson, 2010), aggression (Farmer & Aman, 2011; Fitzpatrick et al., 2016; Mazurek et al., 2013), destructive behavior, and temper tantrums (Esteves et al., 2021; Maskey et al., 2013) and social problems such as poor peer relationship and engagement (Koegel et al., 2001, 2012; Locke et al., 2010), difficulty establishing and maintaining friendships

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(Stewart et al., 2006; Strain & Schwartz, 2001), and loneliness (Bauminger et al., 2003; Jobe & White, 2007) have been noted in individuals with ASD. Given the impact of these difficulties on the overall quality of life for these individuals, it is critical to provide them with proper support (Koldas et al., 2023; Wentzel et al., 2004). The increasing prevalence and severity of ASD highlight the need to establish the most effective therapeutic options for its treatment (Makrygianni et al., 2018).

Interventions based on applied behavior analysis (ABA) principles have been found to be effective in improving the academic, behavioral, and social-emotional performance of individuals with ASD (e.g., Cohen et al., 2006; Dixon et al., 2021; Zachor et al., 2007). For example, implementing a series of functional analyses was effective in reducing destructive behavior and in increasing desired behavior in an adolescent with ASD (Kuhn et al., 2009). Koegel et al. (2012) showed that the social engagement of elementary school children with ASD increased after participating in a behavioral intervention. Vietze and Lax (2020) implemented an ABA-based intervention for toddlers, which increased social-emotional, cognitive, communication, and motor skills and reduced autism-related behavior symptoms. Overall, ABA therapy has shown tremendous benefits for social, cognitive, and language skills for individuals with autism across all age ranges (Ivy & Schreck, 2016; Vietze & Lax, 2020; Virues-Ortega, 2010) and is recognized as the most efficacious practice for these individuals (Fein et al., 2013). Therefore, such interventions have been found to be very appealing to parents, educators, and practitioners because they are implemented systematically and are adaptable to the unique circumstances of students on the spectrum.

Though the implementation of ABA-based practices has a relatively long history in the USA, the first ABA study for children with ASD was conducted in 1999 in Turkey. Birkan (1999) used a single-case experimental design across behaviors to investigate the efficacy of physical and verbal cues for teaching three simple directions (i..e, *stop*, *come here*, *sit here*) to a 4-year-old boy with ASD in a classroom setting. This study was unique because the researcher systematically taught learner readiness skills to a young child on the spectrum in an inclusive classroom environment. An increase in the frequency of occurrence of targeted behaviors was observed, but treatment fidelity was not measured in the study, and there was a lack of social validity data.

Despite a rise in the number of studies over time, the scope of ABA-based therapies and research output remained limited until the last decade. The first review study conducted by Sucuoglu and Demir (2012) included studies between 1990 and 2010, revealing that 14 studies utilized ABA-based interventions in Turkey. In this analysis, the number of published studies was only five (i.e., Akmanoglu & Batu, 2004; Birkan, 1999; Dogru et al., 2007; Tekin-Iftar 2008; Yilmaz et al., 2005) and the remaining studies were unpublished master theses and dissertations. The topics ranged from teaching play skills and motor skills, following simple directions, and pointing to numerals. However, various topics such as academic (e.g., math and reading), behavioral, social, and communication skills were not a focus in any of these studies.

Although several interesting findings were reported in this review study, several limitations were present. First, the effect sizes were calculated for all the studies included in the review, not only those considered to provide sufficient methodological rigor. When effect estimates were generated from research that did not meet methodological criteria, the extent of a practice's effectiveness could be skewed by findings from studies that lacked proper experimental control and measurement reliability (Lory et al., 2020). Second, the effect sizes were measured with the percentage of nonoverlapping data (PND) and the percentage of data points exceeding the baseline phase median (PEM) in this analysis. However, these effect size analyses are not sensitive to trends and lack sensitivity to the magnitude of the treatment effect (Aydin & Tanious, 2021; Vannest & Ninci, 2015). Third, no potential moderating variables for participants, practitioners, or dependent variables were identified to determine whether ABA-based interventions had differential effects based on participant or contextual variables.

In addition to these limitations, Sucuoglu and Demir's (2012) review only included studies up to 2010. Since then, there have been improvements in the research output of ABA-based interventions for individuals with ASD in Turkey, particularly with the contribution of Dr. Elif Tekin-Iftar and her team (e.g., Akmanoglu & Tekin-Iftar, 2011; Kiyak & Tekin-Iftar, 2022; Tekin-İftar & Olcay-Gül, 2016). The scope of these studies was varied compared to previous years with interventions such as picture exchange communication system (PECS; Odluyurt et al., 2016) and pivotal response training (e.g., Besler and Kurt). Additionally, these practices involved parents (e.g., Besler & Kurt, 2016) and teachers (e.g., Tunc-Paftali & Tekin-Iftar, 2021) in relatively more diverse settings such as general education classrooms (e.g., Rakap & Balikci, 2017), special education classrooms (e.g., Ozen et al., 2017), and homes (e.g., Besler & Kurt, 2016).

Given the significance of culture in guiding ABA practices (Dennison et al., 2019; Liao et al., 2020; Sugai et al., 2012), the recent improvements in the implementation of ABA-based interventions in Turkey, and the lack of a recent literature review, a more in-depth investigation of the Turkish research literature is needed that overcomes the aforementioned limitations. In the current meta-analysis, (a) studies that met What Works Clearinghouse (WWC; Kratochwill et al., 2013) design standards (DSs) with and without reservations were included for effect size analyses, (b) two different effect size measurements were included—baseline-corrected tau (Tarlow, 2017) and performance criteria-based

effect size (PCES; Aydin & Tanious, 2021)—that consider performance improvement rather than overlapping data, and (c) moderator variables were identified to provide information about the generalizability of positive effects to subpopulations or contexts.

Given that the most common research design used in the field of ABA is single-case designs (Aydin, 2023; Neil et al., 2021), this study aimed to investigate the prevalence and the effectiveness of ABA-based interventions using single-case experimental designs with individuals with ASD in Turkey, particularly in educational settings. Due to the limitations of the previous analysis (i.e., Sucuoglu & Demir, 2012), we did not limit the review to any time limitations. The specific research questions were the following:

1. What is the current status of ABA-based interventions for individuals with ASD in Turkey?
2. What is the magnitude of the effect of ABA-based interventions used with individuals with ASD in Turkey?
3. Are there any differential effects on age, gender, and comorbidity of ASD with other disabilities based on variables related to students' characteristics?
4. Do intervention characteristics (i.e., dependent variable, independent variable, practitioners, settings, instructional arrangement) moderate the magnitude of change in targeted behaviors?

Method

Search Procedure

A systematic search was conducted in English and Turkish to review ABA-based interventions conducted in Turkey with the local population. The search procedure included five distinct phases (see Fig. 1). The reporting of this meta-analysis was guided by the standards of the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) Statement (Sarkis-Onofre et al., 2021). However, this review was not registered.

Phase 1: Search-I

The following databases were searched for peer-reviewed studies in the English Language Databases: APA PsycNet (i.e., PsycInfo, PsycArticles, PsycThearpy, and PsycBooks), EBSCO, ERIC, ProQuest, and Web of Science in July, 2023. Given that the authors were primarily interested in ABA interventions used in the educational field, specifically in special education departments in Turkey (Sucuoglu & Demir, 2012), the medical databases and medical subject headings were not used in this review. Search terms for ABA-based interventions were determined by the first and second authors looking at recent meta-analysis and systematic review studies for ABA interventions (e.g., Liao et al., 2020; Neil et al., 2021) and revised and confirmed by the third author who is a BCBA-D and a professor in special education with over 30 years of experience. Table 1 demonstrates the search terms for English databases. The first and second authors conducted this search independently on the same databases and transferred the findings to Rayyan. Rayyan is a free web tool designed to help researchers conduct systematic reviews to speed up the selection process of studies and increase the consistency in determining included studies. To calculate the interrater agreement for this phase and also the following phases, a point-by-point method was used to determine the percentage of interrater agreements by dividing the number of agreements by the total number of agreements plus disagreements and multiplying by 100. The interrater agreement was 100% for this phase.

Phase 2: Search-II

In the second phase, the authors conducted a systematic search in Turkish on a Turkish database called Tr Dizin. The search was conducted using the Turkish translation of keywords reviewed for cultural appropriateness by a Turkish native-speaking special education professor with over 25 years of teaching and research experience. However, the key terms "Turk, Turkish, and Turkey" were removed during this phase, given that the database predominantly includes studies conducted in Turkey with the local population. The first and second authors conducted this search independently on the same Turkish database and transferred the findings to Rayyan. The interrater agreement was calculated using the same methods as described in phase 1. After calculating the interrater agreement with a 100% consensus, the second author compiled all articles written in English and Turkish on Rayyan and began collaborating with the first author to review the selected articles for further analysis in his Rayyan account.

Phase 3: Screening and Eligibility

Phase 3 included an additional set of criteria to narrow the list to those articles that were empirical in nature, implemented with the Turkish population, and included participants with ASD. These criteria were as follows: (a) the titles could not include review or meta-analysis, (b) the study had to be conducted in Turkey with the local population, and (c) at least one of the participants had to have a diagnosis of ASD. The first and second authors coded all the articles independently. The interrater agreement was calculated using the same methods as described in phase 1. The interrater agreement was 100% in this phase.

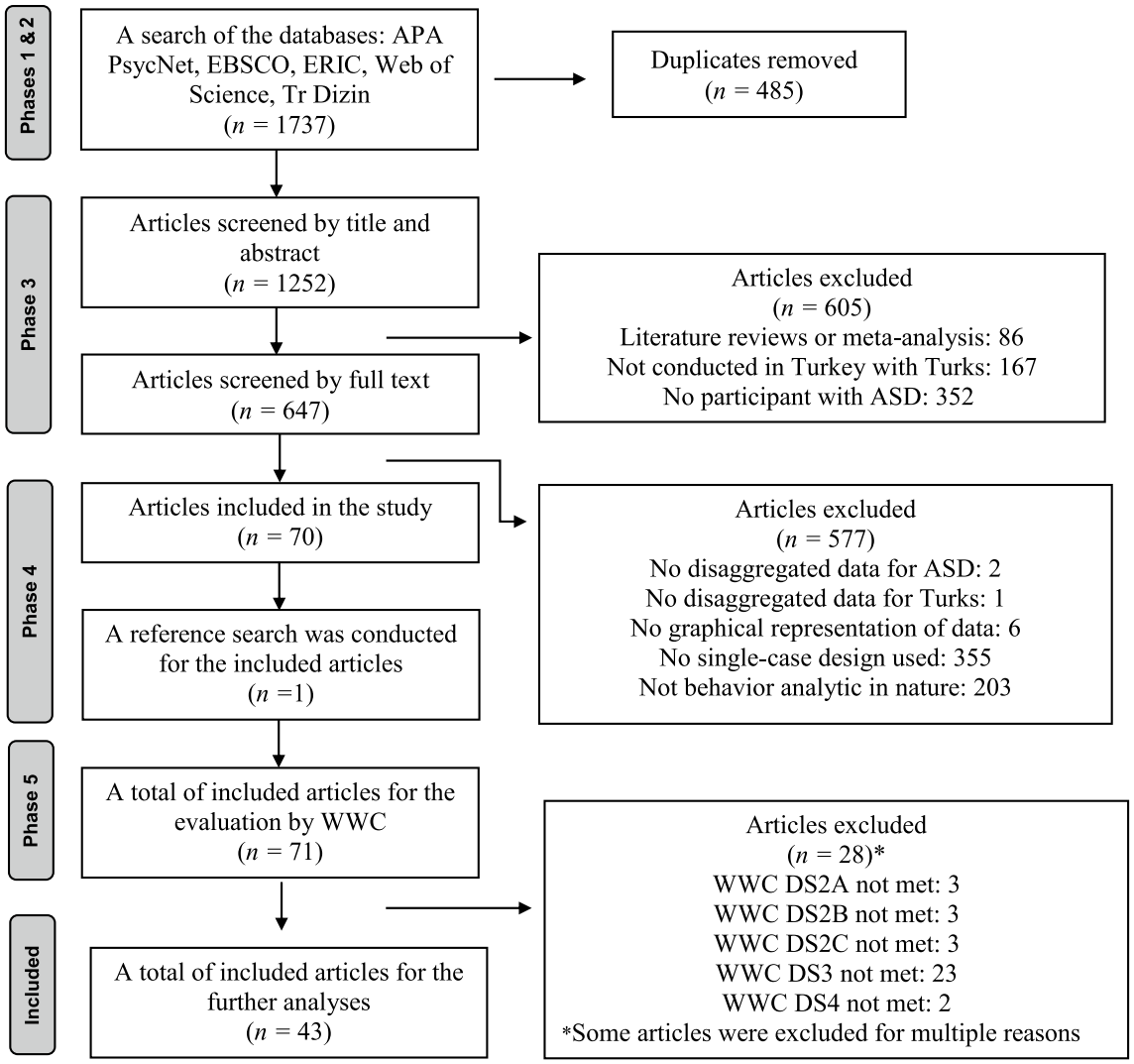


Fig. 1 PRISMA flow diagram. DS, design standard; WWC, What Work Clearinghouse

Phase 4: Inclusion and Exclusion

The fourth phase of the review used the following criteria: (a) at least one of the participants had to be diagnosed with ASD; however, if a participant was part of a group of several others who did not have an ASD diagnosis, the data had to be disaggregated; (b) if the study was conducted collaboratively with non-Turkish institutions including international participants, the data had to be disaggregated for Turkish students with ASD; (c) the study had to utilize a single-case experimental design (SCEDs); (d) there had to be a graphical representation of data in a line format; and (e) the intervention had to be behavior analytic in nature. For a study to be considered behavioral analytic in nature, the following requirements had to be met: (a) the independent variable had to be explicitly operant, (b) the dependent variable had to be sufficiently operationalized, and (c) the results had to be explained based on the principles of ABA as opposed to the use of a mentalistic explanation (Fisher et al., 2013; Neil et al., 2021). The first and second authors coded all the articles independently for interrater agreement in this phase. The interrater agreement was calculated using the same methods as described in phase 1. The interrater agreement was 100% in this phase. The second author also reviewed the reference lists of studies selected for further analyses to be able to capture related research.

Phase 5: Quality Assessment

Studies that met inclusion criteria were further coded for quality evaluation based on WWC DSs (Kratochwill et al., 2013). The included studies were coded according to six WWC DSs as adapted by Maggin et al. (2013) and modified by Hong et al. (2016) (see Table 2). WWC DSs evaluate single-case

Table 1 Search terms for study ıdentification in English language databases

Database	Category	Search terms
EBSCO, ERIC, Apa PsycNet, Web of Science	Autism Turk, Turkish, Turkey	"Austism*" OR "ASD" OR "Asperger" OR "developmental disab*" OR "developmental delay" AND "Turk" OR "Turkish" OR "Turkey" OR "Turkiye" "applied behavior* analysis" OR "functional behavior* assessment*" OR "token economy*" OR "abolishing operation*" OR "ABC checklist" OR "differential reinforce*" OR "abstinence reinforce*" OR "interobserver agreement" OR "schedule* of reinforce*" OR "backward chaining" OR "forward chaining" OR "different* reinforce*" OR "differential reinforce* of alternative behavior" OR "prompting behavior" OR "prompting behavior" OR "continuous reinforce*" OR "direct assessment*" OR "delay of reinforce*" OR "contingent escape" OR "functional analysis*" OR "function-based behavior*" OR "continuous reinforcement schedule*" OR "precision teaching" OR "antecedent intervention*" OR "function based intervention*" OR "discrete trial training" OR "extinction burst" OR "motivating operation*" OR "establishing operation*" OR "extinction schedule*" OR "fading prompt*" OR "functional alternative behavior*" OR "transition assessment*" OR "stimulus" OR "stimulus preference assessment*" OR "positive punishment*" OR "positive reinforce*" OR "negative punishment*" OR "negative reinforce*" OR "ABI" OR "applied behavioral intervention" OR "IBT" OR "intensive behavior intervention" OR "ABAI OR IBT" OR "discrete trial training" OR "behavior therapy" OR "early behavioral treatment" OR "functional behavioral analysis" OR "verbal behavior" OR "self-monitoring" OR "self-recording" or "self-evaluation" OR "self-regulation" OR "self-reporting" OR "self-management" OR "self-reinforcement" OR "self-observation" OR "self-graphing" OR "self-control" OR "self-management" OR "self-regulation" "self-monitoring" OR "self-reinforcement" OR "self-advocacy" OR "self-observation" or "self-instruction" OR "choice-making" OR "decision-making" OR "problem solving" OR "goal setting" OR "attainment" OR "planning" OR "self-advocacy" OR "self-awareness" OR "self- knowledge" or "multicomponent" OR "self-determination" OR "discrete trial teaching" OR "DTT" OR "pivotal response treatment" OR "PRT" OR "picture exchange communication system" OR "PECS" "early start denver model" OR "ESDM" OR "psychotherap*" OR "cognitive behavi* therap*"

Table 2 WWC design standards

DS 1: When the independent variable was systematically manipulated, it is coded with "1." A score of "0" was assigned when the independent variable was not subjected to systematic manipulation

DS 2A: Interobserver agreement is denoted by the code "1." When no interobserver agreement was reported, a rating of "0" was assigned

DS 2B: Interobserver agreement was scored as "2" when at least 20% of all sessions and at least 20% of data points for each condition were provided. When interobserver agreement was observed for at least 20% of all sessions but not for at least 20% of the data points for each condition, a "1" was assigned. Studies were scored as "0" if less than 20% of all sessions had documented interobserver agreement

DS 2C: If interobserver agreement averaged 80% or above, it was coded as "2," the interrater agreement averaged 80% for some but not all conditions, the rating of "1" was given. Studies averaged less than 80% of interrater agreement was coded with "0."

DS 3: When three attempts at three distinct times for ABAB design, multiple baseline designs, and changing criterion design were included to demonstrate intervention effect, it was coded as "2." When four attempts at four distinct times for alternating treatment design were included to demonstrate intervention effect, it was coded as "4." When an alternative treatment design required four attempts and the other designs required three attempts for at least one participant, a coding of "1" was used. If studies using alternative treatment designs featured less than four data points and all other designs included fewer than three data points for at least one participant, they were categorized as "0."

DS 4: When a minimum of five data points was provided for each condition, it was coded as "2." When each condition contained at least three data points, a "1" was coded. Any condition with fewer than three data points was recorded as "0" in studies

Overall rating: When all DSs 1–4 were met with the greatest possible rating, the study was coded "meet standards". If at least one of the DSs 1–4 was coded with "1" where the highest score was 2 and then the study was coded "meet standards with reservations." An overall rating of "0" indicated that at least one DS 1–4 was coded with a "0" and considered "did not meet standards."

DS, design standard

studies regarding (a) systematic manipulation of independent variables, (b) interrater reliability, (c) three demonstrations of the intervention effect, and (d) if an adequate number of data points is provided for each condition. These standards also have been emphasized to determine the risk of bias in SCEDs (cf. Tate et al., 2014). WWC DSs classify studies in three categories: (a) meets without reservation, (b) meets with reservation, and (c) does not meet DSs. The studies were rated as "meet standards (MS)" with a "2" when all standards were met with the highest possible rating. If at least one of the DSs was rated with a "1" where the maximum score was "2," the study was coded as "meet standards with reservations (MS-R)." A score of "0" showed that at least one of the DSs was coded with a "0" and did not meet standards. The first and the second authors analyzed all the studies independently according to WWC DSs. The interrater agreement was calculated using the same methods as described in phase 1. The interrater agreement was 95% in this phase. Note that the authors discussed any disagreements until they reached a 100% consensus. WWC recommends including studies met with and without reservation for further analyses. Therefore, those that met these standards with and without reservations were included for further analyses in the present study.

Descriptive Assessment

The studies that met WWC DSs (i.e., evaluated as MS or MS-R) were coded in a descriptive table created by the first two authors. The descriptive assessment included the following information: (a) participant(s), (b) dependent variables, (c) independent variable, (d) research design, (e) practitioner, (f) setting(s), (g) instructional arrangement, (h) social validity, and (i) mastery criterion. The first and the second authors independently coded all the articles to obtain interrater agreement. The interrater agreement was calculated using the same methods as described in phase 1. Agreement between the first and second authors was 98%. The authors discussed any disagreements until they reached a 100% consensus.

Effect Size Analysis

Several analyses were conducted for the review with two different effect size measurements (i.e., tau, PCES).

Data Extraction

In this review, the PlotDigitizer, a reliable and valid software program for digitizing single-case graphical data (Aydin & Yassikaya, 2022), was used for the data extraction. Data extraction was performed for only data in baseline (A) and intervention (B) phases to allow A-B contrast. The data extraction included only baseline and intervention phases; data points in generalization and maintenance were not extracted for effect size analysis. The digitized data were transferred to the Microsoft Excel file and rounded to the nearest integer value by looking at each graph, as suggested by Aydin and Yassikaya (2022). Effect sizes were calculated using these digitized data. The first author extracted from 43 (100%) studies, and the fifth author extracted data from 23 (53%) randomly selected studies independently. Based on these randomly selected studies, interrater agreement between the first and fifth authors was calculated by accepting $+/-1$ unit deviations as consensus in digitizations and found to be at 94%. The authors discussed any disagreements and looked at the related graphs again to control their data extractions until they reached 100% consensus.

Effect Size Calculations

We calculated baseline-corrected tau (Tarlow, 2017) and PCES (Aydin & Tanious, 2021) for the included studies. We chose these effect sizes because both can control baseline trend which may disrupt the calculation of actual intervention effects, and it is clear when baseline trend should be controlled in both effect sizes. Tau is a nonoverlap-based index and has been used for over a decade in the field, while PCES is a new index based on the mean of data in phases and the level of skill acquisition through intervention. Baseline corrected tau (BC-tau) (Tarlow, 2017) is a form of tau analysis to improve removing monotonic baseline trends from the intervention data. BC-tau should be used if baseline data has a trend at the level of statistical significance; otherwise, tau should be used. On the other hand, PCES methods calculate the effect size of the magnitude of behavior change according to the degree of skill acquisition with the introduction of treatment. PCES values can be calculated according to the mastery criterion set in the studies, and thus effect size of an intervention can be determined based on its ability to approach the mastery criterion. PCES methods include PCES without baseline trend (PCES), PCES with baseline trend ($PCES_{trend}$), and PCES with immediate effect ($PCES_{immediate}$). In this study, PCES with and without baseline trend values (PCES and $PCES_{trend}$) were calculated. The split-middle method determines whether a baseline trend conceals treatment performance in a single-case method. PCES and $PCES_{trend}$ values were calculated by hand for the studies set to a mastery criterion according to the formulas suggested by Aydin and Tanious (2021), whereas tau values were calculated via a web-based calculation engine (https://ktarlow.com/stats/tau/). Moreover, unweighted average tau/BC-tau and PCES/$PCES_{trend}$ values were calculated by hand to determine the overall effect of the studies and moderators.

There are no clear benchmarks for interpreting tau effect sizes. However, tau values can be interpreted small for 0.20, moderate/medium for 0.21–0.60, large for 0.61–0.80, and very large for 0.81 or above (Vannest & Ninci, 2015). In this study, we followed interpretations of tau effect sizes suggested by Vannest and Ninci (2015). On the other hand, there are clear

benchmarks for interpreting PCES and $PCES_{trend}$ values separately (cf. Aydin & Tanious, 2021). PCES interpretations were made based on the more calculated PCES values (PCES or $PCES_{trend}$) for the graphs of the studies. If $PCES_{trend}$ was calculated more in a study or a moderator, the following interpretations were used for the average effect sizes: $PCES_{trend}$ is interpreted ineffective for 0.20 and below, very small effect for 0.21–0.39, small effect for 0.40–0.67, moderate effect for 0.68–0.91, effective for 0.92–1.11, high effect for 1.22 and above. If PCES was calculated in a study or a moderator, the following interpretations were used for the average effect sizes: ineffective for 0.39 and below, very small effect for 0.40–0.60, small effect for 0.61–0.84, moderate effect for 0.85–1.01, effective for 1.02–1.16, and high effect for 1.17 and above. The interrater agreement was calculated using the same methods as described in phase 1. The first and the second authors calculated the effect size of 43 (100%) studies, and interrater agreement was 100%.

Moderator Variable Analyses

Moderator variable analyses for single-case meta-analysis studies could help uncover the boundaries of interventions by evaluating variables such as school levels, types of disability, and varieties of independent and dependent variables separately in a categorical unit (Dowdy et al., 2021; Ledford & Gast, 2018). For the present study, moderators were determined according to the salient variables in the studies. The salient variables were determined in accordance with the descriptive coding of the included articles. As a result, the moderator variables were separated into participant and intervention characteristics. The moderator variables were coded by the first two authors independently for all the studies. The interrater agreement was calculated using the same methods as described in phase 1. The interrater agreement was 96% across the authors. The authors discussed any disagreements until they reached 100% consensus.

Participant Characteristics

Variables of interest were based on participant characteristics and included age group, gender, and additional disability comorbidity. Gender was coded female and male. Age group (i.e., preschool, elementary, secondary) and additional disability comorbidity (i.e., ASD, ASD + ID, ASD + other) were divided into different levels. If any information was not provided for one of the characteristics in an article, we omitted this article for the interest moderator analysis.

Intervention Characteristics

Variables related to intervention characteristics were coded based on (a) dependent variables including academic, communication, daily living, motor, play, and social skills; (b) independent variables including errorless teaching, video-based instructions, pivotal response training, social stories, and combined (including multiple interventions) instructions; (c) practitioners including teachers, researchers, and parents; (d) settings including individualized instructional setting, special education classroom, general education classroom, homes, gym/swimming pools, and outside (i.e., cafeteria, park); and (e) instructional arrangement including one on one and small group. If there was no information for one of these characteristics in a study, we excluded this study for the interest moderator analysis.

Effect Size Comparisons

To compare the effect sizes between categories within a variable, three different statistical analyses were used. First, if there were only two categories, such as gender (i.e., male and female), the Mann–Whitney U test (Mann & Whitney, 1947) was employed. The Mann–Whitney U test is a nonparametric test used to compare two independent variables. Second, for the variables that have more than two categories, such as age (i.e., preschool, elementary, and secondary), the Kruskal–Wallis test statistical analysis (Kruskal & Wallis, 1952) was employed. The Kruskal–Wallis is a nonparametric approach to compare three or more groups to measure at least one ordinal level. Finally, the Dunn post hoc test (Dunnett's T3; Dunn, 1964) was employed if there was a significant difference among the groups based on the Mann–Whitney U and the Kruskal–Wallis test statistical analyses.

Results

Search Procedure

Based on the search procedure in two languages and addressing the inclusion and exclusion criteria, 43 studies were further evaluated for effect size analysis (see Fig. 1). The result of each phase is delineated below.

Phase 1: Search I

A total of 895 studies were identified for this phase based on the review conducted in July, 2023, across all the databases. Even though the search was conducted in English for this phase, the electronic search yielded Turkish written studies, which were included for further review.

Phase 2: Search II

A total of 842 studies were identified in the Turkish database. Of 1737 studies found in phases 1 and 2, duplicate

articles were removed ($n = 485$), resulting in 1252 articles for potential inclusion in this review.

Phase 3: Screening and Eligibility

Two researchers concluded that 605 studies were excluded in this phase for the following reasons: Eighty-six were literature reviews or meta-analyses, 167 studies were not conducted in Turkey with the local population, and 352 studies did not have participants with ASD. After implementing the criterion in phase 3, 647 articles were yielded as potential inclusion candidates.

Phase 4: Inclusion and Exclusion

A total of 577 studies were excluded in this phase for the following reasons: Two studies did not disaggregate data for participants with ASD, one study did not disaggregate data for Turkish students with ASD, six studies did not include the graphical representation of data, 355 studies did not use single-case research design, and 203 studies were not behavior analytic in nature. A total of 70 studies were included for further review. For the included studies, a reference search was conducted, which yielded one new study, bringing the total to 71 articles included at this stage.

Phase 5: Quality Assessment

Following the application of the WWC DSs (see Table 3), 43 (60%) of the 71 articles were included in further analyses. Two studies (3%) met standards without reservations, 41 (56%) studies met standards with reservations, and 28 (40%) studies did not meet standards. Among the studies that did not meet standards, DS1 was met across all the studies. Three studies (4%) failed to meet DS 2A, 2B, and 2C. Twenty-three studies (32%) did not meet DS3. Two studies (3%) did not meet DS4. The 41 (56%) studies meeting standards with reservations were coded with the highest scores for DS1, DS 2A, 2B, 2C, and DS3. However, they were coded with a "1" for DS4, where the highest score was "2."

Descriptive Assessment

Table 4 depicts the result of descriptive and effect size analysis. The 43 studies meeting WWC DSs with and without reservations are listed in the table.

Participant(s)

A total of 127 participants with ASD were included across the 43 studies. Of the 127 participants, 74 were preschool, 45 were elementary, and eight were secondary level. A total of 111 participants were male and a total of 16 participants were female. There were 110 participants who had ASD only, 13 had ASD and ID, three had ASD and blindness, and one had ASD and a speech-language disorder.

Dependent Variables

Dependent variables showed differences across the 43 studies. Seven studies (16.3%) targeted daily living skills such as using a spoon to eat, cooperating with injections, and grocery shopping. Nine studies (20.9%) targeted academic behaviors such as naming numbers, Turkish-art core content, and mathematics (i.e., addition). Six studies (14.0%) targeted motor skills such as riding a bike, swimming, and playing tennis. Seven studies (16.3%) focused on play behaviors such as digital gaming, pretend playing, and building a Lego train. Eight studies (18.6%) targeted social skills, including participating in ongoing activities and naming facial expressions. Finally, six studies (14.0%) targeted communication skills such as naming relatives, manding, and asking questions.

Independent Variables

Seventeen studies (37.2%) used errorless teaching approaches such as prompting procedures and embedded instructions. Eleven studies (25.6%) used video-based instructions. Seven studies (16.3%) employed combined instructions where more than one ABA-based intervention was implemented, such as a combination of a token economy and video modeling. Two studies (4.7%) used pivotal response training and two studies (4.7%) employed social stories. The independent variables of the remaining five studies (11.6%) are PECS, incidental teaching, self-monitoring, direct instruction, and cool vs. not cool.

Research Design

Of the 41 (95.3%) studies that used a multiple-probe design, 25 (61.0%) used a multiple-probe design across participants, 15 (39.0%) used a multiple-probe design across behaviors, and one study (2.4%) used multiple-probe both across participants and across behaviors. Of the 43 studies, one study (2.3%) used a changing criterion design, and one study (2.3%) used an ABACA design.

Practitioner(s)

Thirty-six studies (83.7%) were conducted by researchers. Teachers conducted four studies (9.3%), and three studies (7.0%) were carried out by parents.

Table 3 WWC design standard evaluation

Article citation	Overall	DS#1	DS#2A	DS#2B	DS#2C	DS#3	DS#4
Acar et al. (2017)	0	1	1	2	2	0	1
Akcin, (2013)	0	1	1	2	2	0	1
Akmanoglu (2015)	1	1	1	2	2	2	1
Akmanoglu and Batu (2004)	1	1	1	2	2	2	1
Akmanoglu and Batu (2005)	0	1	1	2	2	0	1
Akmanoglu et al. (2015)	0	1	1	2	2	0	1
Akmanoğlu and Pektas-Karabekir (2020)	1	1	1	2	2	2	1
Akmanoglu and Tekin-Iftar (2011)	1	1	1	2	2	2	1
Akmanoglu et al. (2014)	0	1	1	2	2	0	1
Alp and Akin (2019)	0	1	1	2	2	0	1
Ardiç and Cavkaytar 2014)	0	1	1	2	2	0	1
Arıcan et al. (2018)	0	1	1	2	2	0	1
Aydin and Cavyaktar (2020)	1	1	1	2	2	2	1
Bahcalı and Ozen (2019)	1	1	1	2	2	2	1
Balcik and Ciftci-Tekinarslan (2012)	1	1	1	2	2	2	1
Batu et al. (2014)	0	1	0	0	0	2	1
Besler and Kurt (2016)	1	1	1	2	2	2	1
Birkan (1999)	0	1	0	0	0	1	1
Birkan et al. (2011)	1	1	2	2	2	2	1
Bozkus-Genc and Yucesoy-Ozkan (2021)	2	1	1	2	2	2	2
Cattik and Odluyurt (2017)	1	1	1	2	2	2	1
Ciftci-Tekinarslan and Kanpolat (2012)	1	1	1	2	2	2	1
Cuhadar and Diken (2011)	0	1	1	2	2	2	0
Dalgin-Eyiip and Ulke-Kurkcuoglu (2021)	1	1	1	2	2	2	1
Dargut-Güler and Erdem (2022)	0	1	1	2	2	0	1
Demirtaş and Aykut (2021)	1	1	1	2	2	2	1
Dogru et al. (2007)	1	1	1	2	2	2	1
Elicin and Tunali (2016)	1	1	1	2	2	2	1
Eren et al. (2013)	1	1	1	2	2	2	1
Ergenekon et al. (2014)	0	1	1	2	2	0	1
Genc-Tosun and Kurt (2017a)	1	1	1	2	2	2	1
Genc-Tosun and Kurt (2017b)	0	1	1	2	2	0	1
Gülsöz and Çikili (2018)	0	1	1	2	2	0	1
Horasan and Birkan (2015)	1	1	1	2	2	2	1
Kahveci and Ataman (2017)	0	1	0	0	0	2	1
Kaya and Yucesoy-Ozkan (2022)	0	1	1	2	2	0	1
Kiyak and Tekin-Iftar (2022)	2	1	1	2	2	2	2
Kurnaz and Yanardag (2018)	1	1	1	2	2	2	1
Kurt and Kutlu (2019)	1	1	1	2	2	2	1
Kurt and Tekin-Iftar (2008)	0	1	1	2	2	0	1
Odluyurt et al. (2016)	0	1	1	2	2	0	1
Olcay gul et al. (2019)	1	1	1	2	2	2	1
Olcay-Gul and Vuran (2019)	1	1	1	2	2	2	1
Ozen et al. (2012)	1	1	1	2	2	2	1
Ozen et al. (2017)	1	1	1	2	2	2	1
Ozen et al. (2022)	0	1	1	2	2	0	1
Pektas-Karabekir and Akmanoğlu (2018)	1	1	1	2	2	2	1
Rakap and Balikci (2017)	1	1	1	2	2	2	1
Safak et al. (2018)	1	1	1	2	2	2	1
Şahin and Özen (2021)	0	1	1	2	2	0	1
Sani-Bozkurt and Ozen (2015)	0	1	1	2	2	0	1

Table 3 (continued)

Article citation	Overall	DS#1	DS#2A	DS#2B	DS#2C	DS#3	DS#4
Saral and Ulke-Kurkcuoglu (2022)	1	1	1	2	2	2	1
Tekin-Iftar (2008)	1	1	1	2	2	2	1
Tekin-Iftar and Birkan (2010)	1	1	1	2	2	2	1
Tekin-İftar and Olcay-Gül (2016)	1	1	1	2	2	2	1
Toret et al. (2015)	1	1	1	2	2	2	1
Tunc-Paftali and Tekin-Iftar (2021)	1	1	1	2	2	2	1
Turhan and Vuran (2015)	0	1	1	2	2	0	1
Ulke-Kurkcuoglu (2015)	0	1	1	2	2	0	1
Ulke-Kurkcuoglu et al. (2015)	1	1	1	2	2	2	1
Ulke-Kurkcuoglu and Kircaali-Iftar (2010)	1	1	1	2	2	2	1
Vuran (2008)	0	1	1	2	2	0	1
Yalcin and Akmanoglu (2013)	1	1	1	2	2	2	1
Yanardağ et al. (2011)	1	1	1	2	2	2	1
Yanardag et al. (2013)	1	1	1	2	2	2	1
Yanardag et al. (2015)	1	1	1	2	2	2	1
Yavuz (2019)	0	1	1	2	0	2	1
Yavuz and Şafak (2017)	0	1	1	2	2	0	1
Yilmaz et al. (2005)	1	1	1	2	2	2	1
Yilmaz et al. (2010a)	0	1	1	2	2	2	0
Yilmaz et al. (2010b)	0	1	1	2	2	0	1

DS, design standard

Setting(s)

Twenty-nine studies (67.4%) were conducted in individualized instructional settings, four studies (9.3%) were carried out in a gym or swimming pool, two studies (4.7%) were conducted in special education classrooms, three studies (7.0%) were conducted in participants' homes, three studies (7.0%) were conducted in general education classrooms, and two studies (4.7%) were conducted outside such as parks and cafeterias.

Instructional Arrangement

Thirty-seven studies (86.0%) were delivered in a one-on-one instructional arrangement, and six (14.0%) were conducted in small-group instruction.

Social Validity

Thirty-nine studies (90.7%) provided social validity, and four (9.3%) did not include any assessments for social validity.

Mastery Criterion

Twenty-seven studies (62.8%) had a mastery criterion, and 16 (37.2%) had no mastery criterion.

Effect Size Analysis

Tau values were calculated for 345 tiers (AB phases) from the 43 studies. BC-tau was calculated for only one of them. The ranges of average tau values were from 0.344 ($p = 0.115$, se = 0.266) to 0.977 ($p = 0.023$, se = 0.052). Thirteen studies (30.2%) had a very large effect size, 19 studies (44.2%) had a large effect size, and 11 studies (25.6%) had a medium effect size.

$PCES_{trend}$ was calculated for 33 tiers, while PCES values were calculated for 223 tiers from 27 studies (see Table 4). The ranges of unweighted average PCES values were from 0.40 to 1.15. Since PCES without baseline trend was calculated more in the studies, the non-trend PCES interpretation was considered. Fourteen studies (51.9%) had a small effect, six studies (22.2%) had a moderate effect, five studies (18.5%) had a very small effect, and two (7.4%) were found to be effective.

Moderator Variable Analysis

Table 5 shows moderator variable analyses under two categories. As a first category, participant characteristics involved age, gender, and comorbid disability. The second category—intervention characteristics—included the dependent variable, independent variable, practitioners, settings, and instructional arrangement.

Table 4 Descriptive assessment and effect size analysis

Study	Participant age, gender, diagnosis	Dependent variable(s)	Independent variable(s)	Research design	Practitioner(s)	Setting(s)	Instructional arrangement	Social validity	Mastery criterion	Effect sizes	
										Tau/BC-tau	PCES/PCES-trend
Akmanoglu (2015)	Suna; P; F; PDD Emre; P; M; ASD Aras; P; M; ASD Ege; P; M; ASD	Naming facial expressions	Video modeling	MPDB	Researchers	IIS at a university unit	One on one	Yes	No	0.881 (very large) ($p = 0.037$, se = 0.19)	—
Akmanoglu and Batu (2004)	Omer; S; M; ASD + ID Baris; S; M; ASD + ID Serap; P; F; ASD	Pointing to numbers	Simultaneous prompting	MPDB	Researchers	IIS at private special education school and a university unit	One on one	Yes	No	0.456 (medium) ($p = 0.167$, se = 0.321)	—
Akmanoglu and Pektas-Karabekir (2020)	Bora; P; M; ASD Umit; P; M; ASD Kamil; P; M; ASD	Drawing of a six-part person	Video prompting	MPDP	Researchers	IIS at a university unit	One on one	Yes	Yes (100%)	0.710 (large) ($p = 0.005$, se = 0.228)	0.61 (Small)
Akmanoglu and Tekin-Iftar (2011)	Fatih; E; M; ASD + ID Haluk; E; M; ASD Ozlem; E; F; ASD + ID	Responding to lures of strangers	Video modeling, graduated guidance, and community-based instruction	MPDP	Researchers	IIS at a university unit and different community settings on campus	One on one	Yes	No	0.472 (medium) ($p = 0.081$, se = 0.279)	—
Aydin and Cavyaktar (2020)	Seda; P; F; ASD + ID	Basic math skills	Simultaneous prompting	MPDB	Parents	Home	One on one	Yes	No	0.834 (very large) ($p = 0.25$, se = 0.267)	—
Bahcalı and Ozen (2019)	Mehmet; S; M; ASD + ID	Making a job interview	Video modeling	MPDB	Researchers	IIS at a social living center	One on one	Yes	No	0.779 (large) ($p = 0.11$, se = 257)	—

Table 4 (continued)

Study	Participant age, gender, diagnosis	Dependent variable(s)	Independent variable(s)	Research design	Practitioner(s)	Setting(s)	Instructional arrangement	Social validity	Mastery criterion	Effect sizes	
										Tau/BC-tau	PCES/PCES-trend
Balcik and Ciftci-Tekinarslan (2012)	Burcin; P; F; ASD	Introducing yourself, asking for help, and participating in ongoing activities	Social stories	MPDB	Researchers	IIS at a private special education school	One on one	Yes	No	0.643 (large) ($p = 0.029$, se = 0.325)	—
Besler and Kurt (2016)	Alp; P; M; ASD Deha; P; M; ASD Ege; P; M; ASD	Building a Lego train	Video modeling	MPDP	Parents	Participants' homes	One on one	Yes	Yes (100%)	0.459 (medium) ($p = .198$, se = 0.359)	0.46 (very small)
Birkan et al. (2011)	Osman; E; M; ASD Yasemin; E; F; ASD Ziya; E; M; ASD	Cooperating with injections	Token system and manual guidance	CCD	Teachers	IIS at a private special education school	One on one	No	Yes (changing)	0.887 (very large) ($p = 0.041$, se = 0.202)	0.93 (moderate)
Bozkus-Genc and Yucesoy-Ozkan (2021)	Semih; P; M; ASD Ata; P; M; ASD Yilmaz; P; M; ASD Tolga; P; M; ASD	Asking questions	Pivotal response treatment	MPDP	Researchers	IIS at a university unit	One on one	Yes	No	0.802 (very large) ($p = 0.0$, se = 0.175)	—
Cattik and Odluyurt (2017)	Can; P; M; ASD Berk; P; M; ASD Alp; P; M; ASD Efe; P; M; ASD	Learning digital gaming skills from SMART board and through observation of peers	Graduated-guidance instruction	MPDB	Researchers	IIS at a university unit	Small group instruction	Yes	Yes (100%)	0.813 (very large) ($p = 0.027$, se = 0.277)	0.80 (small)

Table 4 (continued)

Study	Participant age, gender, diagnosis	Dependent variable(s)	Independent variable(s)	Research design	Practitioner(s)	Setting(s)	Instructional arrangement	Social validity	Mastery criterion	Effect sizes	
										Tau/BC-tau	PCES/PCES-trend
Ciftci-Tekinarslan and Kanpolat (2012)	Mehmet; E; M;ASD + ID; Emrah; E; M; ASD + ID; Erdinc; E; M; ASD + ID	Showing clothes	Computer-assisted simultaneous prompting procedure	MPDP	Researchers	ISS at a private special education center	One on one	Yes	No	0.69 (large) ($p = 0.044$, se = 0.340)	—
Dalgin-Eyiip and Ulke-Kurkcuoglu (2021)	Ali; P; M; ASD Omer; P; M; ASD Eda; P; F; ASD Ahmet; E; M; ASD	Following schedules in pretend play skills	Video-enhanced activity schedules	MPDP	Researchers	IIS at a university unit	One on one	Yes	Yes (100%)	0.529 (medium) ($p = 0.66$, se = 0.326)	0.48 (very small)
Demirtaş and Aykut (2021)	P1; P; M; ASD; P2; P; M; ASD; P3; P; M; ASD	Social communication skills	Pivotal response teaching	MBDP	Researchers	IIS at a university unit	One on one	Yes	No	0.588 (medium) ($p = 0.014$, se = 0.251)	—
Dogru et al. (2007)	P1; P; M; ASD P2; P; M; ASD P3; P; M; ASD	Riding balance wheeled bicycle	Direct instruction	MPDP	Researchers	IIS and a park at a private special education rehabilitation center	One on one	No	Yes (5/5)	0.893 (very large) ($p = 0.032$, se = 0.225)	0.89 (moderate)
Elicin and Tunali (2016)	Ali; P; M; ASD Bora; P; M; ASD Cem; E; M; ASD	Schedule following on daily living skills	Graduated guidance with tablet computers	MPDP	Researchers	IIS at a private special education rehabilitation center	One on one	Yes	Yes (100%)	0.645 (large) ($p = 0.048$, se = 0.339)	0.53 (very small)
Eren et al. (2013)	Ege; P; M; ASD; Baran; P; M; ASD Meram; P; F; ASD	Orff-based music activities	Embedded teaching through most to least prompting	MPDP	Researchers	ISS at a private special education rehabilitation center	One on one	Yes	No	0.344 (medium) ($p = 0.115$, se = 0.266)	—
Genc-tosun and Kurt (2017a)	Eren; P; M; ASD Bora; P; M; ASD Ozan; P; M; ASD	Multi-step requesting skills	Discrete trial teaching, time delay, graduated guidance, and reinforcement	MPDP	Researcher	IIS at a university unit	One on one	Yes	Yes (8/10)	0.681 (large) ($p = 0.009$, se = 0.264)	0.69 (small)

Table 4 (continued)

Study	Participant age, gender, diagnosis	Dependent variable(s)	Independent variable(s)	Research design	Practitioner(s)	Setting(s)	Instructional arrangement	Social validity	Mastery criterion	Effect sizes	
										Tau/BC-tau	PCES/PCES-trend
Horasan and Birkan (2015)	Ali; P; M; ASD Hasan; P; M; ASD Gokhan; P; M; ASD	Verbally demanding lost objects	Incidental teaching	MPDP	Researchers	Participants' classroom at a private special education school	One on one	Yes	Yes (8/9)	0.470 (medium) ($p = 0.303$, se = 0.25)	0.54 (very small)
Kiyak and Tekin-Iftar (2022)	P2; S; F; ASD P3; S; M; ASD	Acquiring academic core content in Turkish art	Simultaneous prompting	MPDP	Teachers	School counselor's office and general education classrooms	One on one	Yes	Yes (100%)	0.783 (large) ($p = 0.0$, se = 0.175)	0.71 (Small)
Kurnaz and Yanardag (2018)	Sena; E; F; ASD Ata; E; M; ASD Ugur; E; M; ASD Ersen; E; M; ASD	Acquiring active video game skills	Video self-modeling	MPDP	Researchers	IIS at a university unit	One on one	Yes	Yes (100%)	0.765 (large) ($p = 0.08$, se = 0.295)	0.87 (moderate)
Kurt and Kutlu (2019)	Kerem; E; M; ASD; Ege; P; M; ASD; Cenk; S; M; ASD	Abduction-prevention skills	Social stories	MPDP	Researchers	Streets, roads, parks (outside)	One on one	Yes	No	0.736 (large) ($p = 0.025$, se = 0.285)	—
Odluyurt et al. (2016)	Deniz; P; M; ASD Emir; P; M; ASD Huseyin; P; M; ASD Eren; P; M; ASD Kerim; P; M; ASD Efe; P; M; ASD	Independent communication and maintenance skills	PECS	MPDP	Teachers	Small group instructional setting at a university unit	Small group	Yes	Yes (100%)	0.765 (large) ($p = 0.03$, se = 0.19)	0.83 (small)

Table 4 (continued)

Study	Participant age, gender, diagnosis	Dependent variable(s)	Independent variable(s)	Research design	Practitioner(s)	Setting(s)	Instructional arrangement	Social validity	Mastery criterion	Effect sizes	
										Tau/BC-tau	PCES/PCES-trend
Olcay gul et al. (2019)	Ismail; E; M; ASD Ege; E; M; ASD Kayra; E; M; ASD	Using signs to find restroom in public settings	Video modeling	MPDP	Researchers	ISS at a private special education and rehabilitation setting	One on one	Yes	Yes (100%)	0.559 (medium) ($p = 0.007$, se = 0.252)	0.61 (small)
Olcay-Gul and Vuran (2019)	Okan; E; M; ASD Orkun; E; M; ASD Emre; E; M; ASD	Coping with inappropriate requests from familiar peers and adults	Cool versus not cool	MPDP	Researchers	IIS at an assessment unit of a university	One on one	Yes	Yes (100%)	0.674 (large) ($p = 0.011$, se = 0.282)	0.71 (small)
Ozen et al. (2012)	Osman; E; M; ASD Cemil; E; M; ASD Uzan; E; M; ASD	Sociodramatic play skills	Video modeling and observational learning	MPDB	Researchers	IIS and a built canteen at a university unit	Small group instruction	Yes	Yes (%80)	0.767 (large) ($p = 0.027$, se = 0.293)	1.09 (effective) (one of the nine graphs is $PCES_{trend}$)
Ozen et al. (2017)	Batu; P; M; ASD	Identifying target pictures	Simultaneous prompting and computer assisted instruction	MPDB	Researchers	Self-contained special education classroom	Small group instruction	Yes	Yes (%100)	0.678 (large) ($p = 0.024$, se = 0.305)	0.68 (small) (six of the 12 graphs are $PCES_{trend}$)
Pektas-Karabekir and Akmanoğlu (2018)	Efe; P; M; ASD Uzay; P; M; ASD Eren; P; M; ASD	Responding gestures, and verbal and facial expressions of happiness and anger	Video modeling	MPDP	Researchers	IIS and a group training room at a university unit	One on one	Yes	No	0.673 (large) ($p = 0.016$, se = 0.29)	—

Table 4 (continued)

Study	Participant age, gender, diagnosis	Dependent variable(s)	Independent variable(s)	Research design	Practitioner(s)	Setting(s)	Instructional arrangement	Social validity	Mastery criterion	Effect sizes	
										Tau/BC-tau	PCES/PCES-$_{trend}$
Rakap and Balikci (2017)	P1; P; M; ASD	Using pictures to request, using a spoon to eat, removing coat	Embedded instruction	MPDB	Researchers	General education classroom	One on one	Yes	Yes (80%)	0.638 (large) ($p = 0.004$, se = 0.232)	1.15 (effective)
Safak et al. (2018)	Ayse; S; F; ASD + blind Busra; S; F; ASD + blind Burak; S; M; ASD + blind	Listening comprehension	Multisensory stories induced with constant time delay	MPDP	Researchers	IIS at a private school for students with visual impairments	One on one	Yes	Yes (80%)	0.578 (medium) ($p = 0.139$, se = 0.224)	0.40 (very small) (four of the six graphs are PCES$_{trend}$)
Saral and Ulke-Kurkcuoglu (2022)	Bora; P; M; ASD Can; P; M; ASD Deniz; P; M; ASD	Pretend play behaviors	Least-to-most prompting procedure with contingent imitation	MPDP	Researchers	IIS at a university unit	One on one	Yes	Yes (5 × the highest baseline score)	0.666 (large) ($p = 0.04$, se = 0.331)	0.69 (small) (15 of the 34 graphs are PCES$_{trend}$)
Tekin-Iftar (2008)	Fatih; E; M; ASD + ID Nilay; S; F; ASD + ID	Grocery shopping, ordering rolls in pastry shop, and ordering dry cleaning	Community-based instruction with simultaneous prompting	MPDP	Parents	Participants' homes	One on one	Yes	Yes (100%)	0.713 (large) ($p = 0.087$, se = 0.576)	0.67 (small) (six of the 12 graphs are PCES$_{trend}$)

Table 4 (continued)

Study	Participant age, gender, diagnosis	Dependent variable(s)	Independent variable(s)	Research design	Practitioner(s)	Setting(s)		Instructional arrangement	Social validity	Mastery criterion	Effect sizes	
											Tau/BC-tau	PCES/PCES-trend
Tekin-Iftar and Birkan (2010)	Ogulcan; E; M; ASD Ulas; E; M; ASD Ali; E; M; ASD	Food and drink preparation skills	Progressive time delay, general case training, and observational learning	MPDP, and MPDB	Researchers		Cafeteria	Small group instruction	Yes	Yes (100%)	0.787 (large) ($p = 0.039$, se = 0.182)	0.75 (small)
Tekin-İftar and Olcay-Gül (2016)	Olgun; E; M; ASD Orhan; E; M; ASD + ID Enes; E; M; ASD + ID	Academic skills in science and social studies	Stimulus prompting, instructive feedback, and observational learning stimuli	MPDB	Researcher		General education classroom	Group instructional arrangement	Yes	Yes (100%)	0.860 (very large) ($p = 0.028$, se = 0.194)	0.91 (moderate)
Toret et al. (2015)	P1; E; F; ASD P2; E; M; ASD P3; E; M; ASD	Academic achievement in addition	Self-monitoring interventions	MPDP	Researchers		IIS at a private special education rehab center	One on one	Yes	Yes (100%)	0.543 (medium) ($p = 0.138$, se = 0.319)	0.67 (small) (one of the six graphs is $PCES_{trend}$)
Tunc-Paftali and Tekin-Iftar (2021)	Ali; P; M; ASD Gizem; P; F; PDD NOS Can; P; M; ASD + SPL	Acquisition of discrete target behaviors	Web-based professional development portal (e-coaching)	MPDP	Teachers		IIS and a general education classroom at a public elementary school	One on one	Yes	Yes (100%)	0.806 (very large) ($p = 0.0$, se = 0.155)	0.80 (small)
Ulke-Kurkcuoglu et al. (2015)	Burak; P; M; ASD; Can; P; M; ASD; Omer; P; M; ASD; Alp; P; M; ASD	On-schedule and role-play skills	Computer-assisted activity schedules	MPDP	Researchers		ISS at a university unit	One on one	Yes	No	0.392 (medium) ($p = 0.227$, se = 0.318)	—

Table 4 (continued)

Study	Participant age, gender, diagnosis	Dependent variable(s)	Independent variable(s)	Research design	Practitioner(s)	Setting(s)	Instructional arrangement	Social validity	Mastery criterion	Effect sizes	
										Tau/BC-tau	PCES/PCES-trend
Ulke-Kurkcuoglu and Kircaali-Iftar (2010)	Utku; P; M; ASD; Alp; P; M; ASD; Selim; P; M; ASD; Yavuz P; M; ASD	On-task behaviors	Most-to-least prompting	ABACA	Researchers	IIS at a private special education rehab center	One on one	No	No	0.78 (large) ($p = 0.28$, se = 302)	—
Yalcin and Akmanoglu (2013)	Faruk; E; M; ASD	English words	Simultaneous prompting procedure	MPDB	Researchers	IIS at a private special education rehab center	One on one	Yes	No	0.86 (very large) ($p = 0.58$, se = 0.269)	—
Yanardağ et al. (2011)	P1; E; M; ASD P2; E; M; ASD P3; E; M; ASD P4; E; M; ASD	Basic tennis skills	Least to most prompting procedure	MPDB	Researchers	Indoor gym	One on one	No	No	0.758 (large) ($p = 0.014$, se = 0.213)	—
Yanardag et al. (2013)	Berat; E; M; ASD Sacit; P; M; ASD Sumru; P; F; ASD	Aquatic play skills	Video prompting	MPDB	Researchers	Indoor swimming pool	One on one	Yes	Yes (100%)	0.862 (very large) ($p = 0.050$, se = 0.264)	0.86 (moderate)
Yanardag et al. (2015)	Sacit; P; M; ASD Erdal; P; M; ASD Emek; P; M; ASD	Advanced moving exploration skills	Most to least prompting procedure	MPDB	Researchers	Indoor swimming pool	One on one	Yes	Yes (100%)	0.956 (very large) ($p = 0.064$, se = 0.072)	0.95 (moderate)
Yilmaz et al. (2005)	Omer; E; M; ASD Ali; E; M; ASD Yener; E; M; ASD Can; E; M; ASD	Aquatic play skills	Constant time delay procedure	MPDB	Researchers	Indoor swimming pool	One on one	No	Yes (100%)	0.977 (very large) ($p = 0.023$, se = 0.052)	0.77 (moderate)

ASD, autism spectrum disorder; *CCD*, changing criterion design; *E*, elementary; *F*, female; *ID*, intellectual disability; *IIS*, individualized instructional settings; *MPDB*, multiple probe design across behaviors; *MPDP*, multiple probe design across participants; *P*, preschool; *P1*, participant 1; *P2*, participant 2; *P3*, participant 3; *P4*, participant 4; *PDD*, pervasive developmental disorders; *PDD NOS*, pervasive developmental disorder–not otherwise specified; *S*, secondary; *SPL*, speak language disorder

Table 5 Effect size measurements for moderators

Variable categories	Number of studies	Number of study participants	Unweighted average tau/BC-tau	Unweighted average PCES/$PCES_{trend}$
Participant characteristics				
Age				
Preschool	26	74	0.699 (large) ($p = 0.055$, se = 0.245)	0.82 (small) (for 44 participants)
Elementary	17	45	0.736 (large) ($p = 0.04$, se = 0.255)	0.77 (small) (for 33 participants)
Secondary	5	8	0.643 (large) ($p = 0.094$, se = 0.262)	0.60 (very small) (for six participants)
Gender				
Female	12	16	0.692 (large) ($p = 0.047$, se = 0.265)	0.78 (small) (for nine participants)
Male	42	111	0.708 (large) ($p = 0.053$, se = 0.248)	0.69 (small) (for 74 participants)
Disability comorbidity				
ASD	39	110	0.712 (large) ($p = 0.05$, se = 0.242)	0.78 (small) (for 75 participants)
ASD + ID	7	13	0.680 (large) ($p = 0.056$, se = 0.344)	0.79 (small) (for four participants)
ASD + other	2	4	0.619 (large) ($p = 0.104$, se = 0.247)	0.55 (very small) (for four participants
Intervention characteristics				
Dependent variable				
Academic	9	21	0.699 (large) ($p = 0.165$, se = 0.254)	0.67 (small) (for 12 participants)
Communication	6	22	0.708 (large) ($p = 0.068$, se = 0.485)	0.72 (small) (for 18 participants)
Daily living	7	18	0.702 (large) ($p = 0.109$, se = 0.297)	0.77 (small) (for 11 participants)
Motor	6	20	0.859 (very large) ($p = 0.031$, se = 0 .175)	0.80 (small) (for nine participants)
Play	7	25	0.684 (large) ($p = 0.172$, se = 0.313)	0.73 (small) (for 21 participants)
Social	8	21	0.668 (large) ($p = 0.034$, se = 0.273)	0.71 (small) (for 12 participants)
Independent variable				
Errorless teaching	17	46	0.731 (large) ($p = 0.049$, se = 0.243)	0.84 (small) (for 24 participants)
Video based	11	35	0.691 (large) ($p = 0.12$, se = 0.277)	0.71 (small) (for 24 participants)
Pivotal response	2	6	0.73 (large) ($p = 0.007$, se = 0.213)	-
Social stories	2	4	0.689 (large) ($p = 0.027$, se = 0.305)	-
Combined	6	16	0.702 (large) ($p = 0.038$, se = 0.246)	0.76 (small) (for seven participants)
Practitioner (s)				
Researchers	35	103	0.704 (large) ($p = 0.075$, se = 0.249)	0.74 (small) (for 63 participants)
Teachers	4	14	0.81 (very large) ($p = 0.017$, se = 0.18)	0.82 (small) (for 14 participants)
Parents	3	6	0.627 (large) ($p = 0.142$, se = 0.467)	0.57 (very small) (for five participants)

Table 5 (continued)

Variable categories	Number of studies	Number of study participants	Unweighted average tau/BC-tau	Unweighted average PCES/$PCES_{trend}$
Settings				
Individualized	28	86	0.680 (large) ($p=0.075$, se = 0.266)	0.76 (small) (for 55 participants)
Gym/swimming pool	4	14	0.888 (very large) ($p=0.037$, se = 0.15)	0.86 (moderate) (for 10 participants)
Sped classroom	2	4	0.574 (moderate) ($p=0.163$, se = 0.277)	0.61 (small) (for four participants)
Home	3	6	0.627 (large) ($p=0.142$, se = 0.467)	0.57 (very small) (for five participants)
Gened classroom	3	5	0.76 (large) ($p=0.01$, se = 0.2)	0.92 (moderate) (for five participants)
Outside	2	6	0.76 (large) ($p=0.035$, se = 0.185)	0.75 (small) (for three participants)
Instructional arrangement				
One-on-one	37	107	0.692 (large) ($p=0.081$, se = 0.257)	0.64 (small) (for 63 participants)
Small group	6	20	0.778 (large) ($p=0.029$, se = 0.249)	0.84 (small) (for 20 participants)

Gened, general education; *Sped*, special education

Participant Characteristics

Age Age was categorized into three school levels—preschool, elementary, and secondary (see Table 5). Of the participants, 74 were preschool age and the overall tau effect size was 0.699 ($p=0.055$, se = 0.245), whereas the overall PCES was 0.82. Forty-five students were of elementary school age, and the overall tau effect size was 0.736 ($p=0.04$, se = 0.255), whereas the overall PCES was 0.77. Eight students were of secondary school age. The overall effect size was 0.643 ($p=0.094$, se = 0.262), whereas PCES was 0.60. All three age categories had a large effect on tau effect size. On the other hand, preschool and elementary students had a small effect while secondary students had a very small effect according to PCES values. As shown in Table 6, Kruskal–Wallis tests for effect sizes indicated no statistically significant difference between participants on age groups ($p=0.208$ for tau and $p=0.331$ for PCES).

Gender Moderator effect sizes were calculated for female students ($n=16$) as 0.692 ($p=0.047$, se = 0.265) and 0.78; moderator effect sizes for male students ($n=111$) were 0.708 ($p=0.053$, se = 0.248) and 0.69 according to tau and PCES, respectively. Both female and male students had a large effect, according to tau, whereas they had a small effect, according to PCES. Mann–Whitney *U* tests for effect sizes

Table 6 Effect size comparisons of moderators

Variable categories	Number of comparison groups	Kruskal–Wallis/Mann–Whitney U (*p*-value)	
		Tau/BC-tau	PCES/$PCES_{trend}$
Age	3	0.208	0.331
Gender	2	0.651	0.608
Disability comorbidity	3	0.177	0.222
Dependent variables	6	0.163	0.144
Independent variables	5	0.807	0.583
Practitioners	3	0.168	0.089
Settings	6	0.092	0.149
Instructional arrangements	2	0.155	0.257

Note: Dunn post hoc test is not performed because comparison of all sub-categories among each other was not statistically significant

indicated no statistically significant difference between participants on gender (see Table 6; $p=0.651$ for tau, $p=0.608$ for PCES).

Disability Comorbidity Disability comorbidity was categorized into three disability categories—ASD, ASD and ID, and ASD and "other" (see Table 5). Most of the students ($n=110$) were only diagnosed with ASD, and their overall tau effect size was 0.712 ($p=0.05$, se = 0.242); the overall PCES was 0.78. Some students ($n=13$) were diagnosed with ASD and ID together and their overall tau effect size was 0.680 ($p=0.056$, se = 0.344); the overall PCES was 0.79. Other students ($n=4$) were diagnosed with ASD and "other" (e.g., blind) and their overall tau effect size was 0.619 ($p=0.104$, se = 0.247) with an overall PCES of 0.55. All disability categories had a large effect for tau values. On the other hand, students with ASD or ASD + ID and students with ASD + other (e.g., blind) had a small effect for PCES. Kruskal–Wallis tests for the effect sizes indicated no statistically significant difference between students on disability comorbidity (see Table 6; $p=0.177$ for Tau, $p=0.222$ for PCES).

Intervention Characteristics

Dependent Variables The dependent variables were studied in six skill categories—academic, communication, daily living, motor, play, and social (see Table 5). Nine studies (20.9%) focused on academic skills for 21 participants and the overall tau effect size was 0.699 ($p=0.165$, se = 0.254), while the overall PCES was 0.67. Eight studies (18.6%) conducted social skills training for 21 participants and their overall tau effect size was 0.668 ($p=0.034$, se = 0.273), whereas the overall PCES was 0.71. Six studies (14.0%) targeted communication skills for 22 participants and the overall tau effect size was 0.708 ($p=0.068$, se = 0.485), while the overall PCES was 0.72. Seven studies (16.3%) focused on play skills for 25 participants and the overall tau effect size was 0.684 ($p=0.172$, se = 0.313), while the overall PCES was 0.73. Seven studies (16.3%) taught daily living skills for 18 participants and the overall tau effect size was 0.702 ($p=0.109$, se = 0.297), while the overall PCES was 0.77. Six studies (14.0%) taught motor skills for 20 participants and the overall tau effect size was 0.859 ($p=0.031$, se = 0.175), while the overall PCES was 0.80. While interventions on motor skills teaching had a very large effect, other interventions had a large effect according to tau interpretation. According to PCES interpretation, all dependent variable categories had a small effect. Kruskal–Wallis tests for effect sizes indicated no statistically significant difference between studies on the dependent variables (see Table 6; $p=0.163$ for tau and $p=0.144$ for PCES).

Independent Variables The independent variables were studied in five instructional categories—errorless teaching, video-based instructions, pivotal response treatments, social stories, and combined (see Table 5). However, five studies (13.9%; Balcik and Ciftci-Tekinarslan 2012; Dogru et al., 2007; Horasan & Birkan, 2015; Olcay-Gul & Vuran, 2019; Toret et al., 2015) were not classified in any of these categories given that each study was the only representation for that group of intervention in Turkey (i.e., PECS, incidental teaching, self-monitoring, direct instruction, and cool vs. not cool). Of the remaining 38 studies, 17 (45.9%) employed errorless teaching methods for 46 participants; the overall tau effect size of them was 0.731 ($p=0.049$, se = 0.243), and the overall PCES was 0.84. Eleven studies (29.7%) used video-based instructions for 35 participants; the overall tau effect size was 0.691 ($p=0.12$, se = 0.277) and the overall PCES was 0.71. Six studies (16.2%) conducted combined instructional methods (e.g., video-model + graduated guidance) for 16 participants; the overall tau effect size was 0.702 ($p=0.038$, se = 0.246) and the overall PCES was 0.76. Two studies (5.4%) used pivotal response treatment for 6 participants and social stories for four participants, respectively (see Table 5). All independent variable categories had a large effect or a small effect according to tau and PCES indicators, respectively. The five studies not included in any of these five categories were considered based on individual effect sizes (see Table 4). The Kruskal–Wallis test indicated no statistically significant difference between studies on the independent variables (see Table 6; $p=0.807$ for tau, $p=0.583$ for PCES).

Practitioners The practitioners were divided into three categories—researchers, teachers, and parents. Only one study (Bozkus-Genc & Yucesoy-Ozkan, 2021) was not included in these categories because both researchers and parents were employed as a practitioner in the study. The interventions in 35 studies (81.4%) were conducted by researchers and involved 103 participants; the overall tau effect size was 0.704 ($p=0.075$, se = 0.249) and the overall PCES was 0.74, which were interpreted as having a large effect and small effect according to tau and PCES interpretations, respectively. The interventions in four studies (9.3%) involving 14 participants were implemented by teachers; the overall tau effect size was 0.81 ($p=0.017$, se = 0.18) and the overall PCES was 0.82, which were interpreted as a very large effect and small effect according to tau and PCES, respectively. The interventions of only three studies (7.0%) with six participants were provided by parents; the overall tau effect size was moderate at 0.627 ($p=0.142$, se = 0.467) and the overall PCES effect size was very small at 0.57. The Kruskal–Wallis test indicated no statistically significant difference between studies on practitioners (see Table 6; $p=0.168$ for tau and $p=0.089$ for PCES).

Settings Settings were categorized into six different learning environments—individualized instructional setting,

gym/swimming pool, special education classroom, home, general education classrooms, and outside. Twenty-eight studies (65.1%) were conducted in an individualized instructional setting with 86 participants; the overall tau effect size was large at 0.680 ($p = 0.075$, se = 0.266) and the overall PCES was small at 0.76. Four studies (9.3%) were conducted at a gym/swimming pool with 14 participants; the overall tau effect size was very large at 0.888 ($p = 0.037$, se = 0.15) and the overall PCES was moderate at 0.86. Three studies (7.0%) were conducted in the home with six participants; the overall tau effect size was large at 0.627 ($p = 0.142$, se = 0.467), and the overall PCES was small at 0.57. Three studies (7.0%) were conducted in general education classrooms with five participants; the overall tau effect size was very large at 0.76 ($p = 0.01$, se = 0.2) and the overall PCES was moderate at 0.92. Two studies (4.7%) were conducted in a special education classroom with four participants; the overall tau effect size was moderate at 0.574 ($p = 0.163$, se = 0.277) and the overall PCES was small at 0.61. Two studies (4.7%) were in outside with six participants, the overall tau was large at 0.76 ($p = 0.035$, se = 0.185), and overall PCES was small at 0.75. Kruskal–Wallis tests indicated no statistically significant difference between studies on settings (see Table 6; $p = 0.092$ for tau, $p = 0.149$ for PCES). Note that one study (Odluyurt et al., 2016) was not included in one of the six categories because this study was the only representation of that setting in Turkey (i.e., a small group setting in a university [Odluyurt et al., 2016]). Therefore, this study was considered with its individual effect size (see Table 4).

Instructional Arrangement Two categories were developed for instructional arrangement—one on one and small group. A total of 107 participants were studied one on one in 37 studies (86.0%), the overall tau effect size was 0.692 ($p = 0.081$, se = 0.257), and the overall PCES was 0.64. These results were interpreted as a large effect for tau and a small effect for PCES. Twenty participants in six studies (14.0%) received small-group instruction; the overall tau effect size was 0.778 ($p = 0.029$, se = 0.249), and the overall PCES was 0.84. These effects were interpreted as a large effect for tau and a small effect for PCES. Mann–Whitney *U* tests indicated no statistically significant difference between studies on instructional arrangements (see Table 6; $p = 0.155$ for tau, $p = 0.257$ for PCES).

Discussion

The purposes of this meta-analysis were to examine the prevalence and the effects of ABA-based interventions conducted with individuals with ASD in Turkey and to determine whether specific participant and intervention characteristics moderated such effects. To do this, we first conducted a systematic search in education and psychology databases, given the usage of ABA-based interventions in educational settings in Turkey (Sucuoglu & Demir, 2012). This search yielded 71 studies. We included only studies ($n = 43$) that met WWC (Kratochwill et al., 2013) DSs with and without reservations to increase confidence in the findings by including methodologically rigorous studies. The analysis occurred in three steps. First, the 43 studies were coded descriptively. Second, individual effect size analysis was conducted with tau. Further PCES analysis (Aydin & Tanious, 2021) was conducted for studies ($n = 27$) that had a mastery criterion. Finally, moderator variables (i.e., participant and intervention characteristics) were compared based on the salient variables derived from the descriptive analysis of the 43 studies.

There was a large increase in the number of ABA-based studies published involving individuals with ASD in Turkey. Although Sucuoglu and Demir's (2012) study included 14 studies (encompassing unpublished theses and dissertations), we had a total of 71 published studies in peer-reviewed journals. Our search might have yielded more studies if we did not omit the gray literature. Alongside the increase in the study volume, there were several important findings based on descriptive and effect size analyses. First, targeted behaviors were diversified with the increased number of studies published. The recent ABA literature included daily living skills, academic behaviors, motor skills, play behaviors, social skills, and communication. The most commonly used teaching strategy was the errorless teaching strategy in both reviews (i.e., 10 of 14 studies in Sucuoglu and Demir (2012) and 17 of 43 studies in the current review). Furthermore, social validity findings were absent in eight of the 14 studies in the previous review (i.e., Sucuoglu & Demir, 2012). In the current review, however, only four studies did not include social validity, two of which (i.e., Dogru et al., 2007; Yilmaz et al. 2005) were also reviewed by Sucuoglu and Demir (2012).

Second, a large magnitude of the overall tau effect size of 0.71($p = 0.072$, se = 0.254) was found in the analysis of the 43 studies included. These findings align with the Sucuoglu and Demir (2012) review (0.85 PND, 90 PEM) despite the different effect size analyses used across the two studies. Both participant and the intervention characteristics generally exhibited large effects as determined by the tau analysis. Conversely, the PCES analysis revealed a small overall effect size of 0.73. The findings for participants and the intervention characteristics were also found to be small. The discrepancies between findings across two different effect size analyses (i.e., PCES, tau) can be attributed to the differing analytical approaches in single-case research. The key distinction here is that PCES focuses on performance criteria, while tau is based on nonoverlap-based effect size measures. Additionally, although our tau effect size can be considered large, it is essential to recognize that benchmarks

for single-case design effect sizes are not as well established as those for group research designs. PCES, however, offered clear benchmarks to interpret the findings. It is also worth noting that none of the sub-categories was statistically significant based on Kruskal–Wallis and Mann–Whitney U tests in both analyses (i.e., tau, PCES).

Regarding the PCES analysis, which revealed a small overall effect, it is important to interpret these findings in light of the predetermined mastery criteria applied across all the studies reviewed. Aydin and Tanious (2021) proposed an acceptable mastery criterion of 80% of the targeted mastery criterion in a study. For example, if the study proposed a 90% mastery criterion, PCES effect size analysis could be conducted at 0.80 of the 90%, which would be 72%. Thus, the size of the current findings could have been greater if the analyses were calculated on acceptable criteria. Sixteen (37.2%) studies were omitted from PCES analyses due to the requirement for a mastery criterion. Despite the nuances in effect size interpretation, our study underscores the growing popularity and potential effectiveness of ABA-based interventions with Turkish individuals with ASD.

Even though we can reach a similar conclusion regarding the overall effectiveness of ABA-based interventions as other meta-analytic reviews in general, a direct comparison of the findings may not be possible due to differences in inclusion criteria, effect size measurements, and methodological flaws (Makrygianni et al., 2018). To our best knowledge, this study is the first meta-analysis conducted to investigate the use of ABA studies specific to a country, certainly outside of the USA. Although there have been multiple meta-analyses for ABA practices, they targeted a specific independent (e.g., early childhood interventions, Virues-Ortega, 2010) or dependent variable (e.g., challenging behaviors, Lory et al., 2020) or were based on specific participant characteristics such as school-based interventions for adults with ASD (David et al., 2022).

Third, the findings of the current review showed that age, gender, and disability comorbidity did not moderate the intervention outcomes across participants based on Kruskal–Wallis and Mann–Whitney U tests. While tau analyses revealed a large effect size for age, gender, and disability comorbidity, including all subcategories, PCES analyses indicated a very small effect size for two subcategories, including secondary and ASD + other, and a small effect size for the remaining subcategories. Our findings based on tau align with the literature that ABA-based interventions were effective in improving behavioral outcomes of individuals regardless of age, gender, and comorbid disability (David et al., 2022; Lory et al., 2020; Neil et al., 2021). The alignment of findings across different cultures suggests that ABA-based interventions may be equally suitable for individuals with ASD in Turkish culture as they are in western cultures, aligning with previous research (e.g., Makrygianni et al., 2018; Roth et al., 2014). These results emphasize the relevance and applicability of ABA principles in diverse cultural contexts, which could have significant implications for the development and implementation of interventions for individuals with ASD worldwide.

Intervention characteristics were placed into five categories including dependent variables, independent variables, practitioners, settings, and instructional arrangement. None of the categories moderated the intervention outcomes across participants based on Kruskal–Wallis and Mann–Whitney U tests. The analyses based on tau suggest a very large effect on motor skills, teachers, and gyms. While special education classrooms and homes had a medium effect, all the remaining subcategories appeared to have a large effect size. Therefore, our findings based on tau align with the previous literature that ABA-based interventions effectively improve the behavioral outcomes of individuals with ASD (Eldevik et al., 2009; Neil et al., 2021). On the other hand, PCES revealed a small effect size for all subcategories but parents, homes, gyms, and general education classrooms. While parents and homes moderated a very small effect size, gyms and general education classrooms moderated a medium effect size.

Even though the findings yielded opposite results across the two effect size analyses, two moderator variables—*parents* and *homes*—seem to have a worse overall moderating effect than any other subcategories within both analyses. Parent-implemented ABA-based interventions effectively improved the behavioral outcomes of individuals with ASD in the previous reviews (i.e., McConachie & Diggle, 2007; Meadan et al., 2009; Patterson et al., 2012; Postorino et al., 2017). Though cultures were not specific consideration in these reviews, the studies reviewed were predominantly implemented in western countries. Given the importance of parents for skill acquisition and maintaining behavioral outcomes, future studies with high methodological rigor are warranted to investigate such effects within the Turkish community. Additionally, only three studies (i.e., Besler & Kurt, 2016; Tekin-Iftar, 2008) were analyzed for these subcategories; therefore, these findings should be regarded with caution.

Although the present meta-analysis review provided support for behavior-analytic therapies for individuals with ASD in Turkey, there were several limitations. First, we did not include gray literature (e.g., theses, dissertations). The exclusion of gray literature may have resulted in publication bias in that unpublished studies may have had small effect sizes (Polanin et al., 2016). A second limitation of the educational aspect of our review pertains to our database selection and search terms. By primarily searching education and psychology databases and using educational terminology, we might have missed relevant studies in medical databases that

could provide a broader perspective on ABA interventions for individuals with ASD. This limitation may restrict the generalizability of our findings, especially when considering the broader healthcare context and potential therapeutic approaches beyond the educational setting. In addition, the exclusive focus on the educational aspect of ABA-based interventions has wider implications for the interpretation of our results. Individuals with ASD often receive services in various domains, including clinical, home, and community settings (e.g., Hernandez & Ikkanda, 2011). Therefore, even though our findings offer valuable insights into the educational setting, they may not fully represent the effectiveness of behavior-analytic therapies in clinical, home-based, or community contexts. This limitation highlights the need for future research to explore the applicability and efficacy of ABA-based interventions across diverse settings to provide a more holistic understanding of their impact on individuals with ASD. Finally, we did not include maintenance or generalization data in our analyses, so we were unable to address interpretations concerning the maintenance or generalization of intervention effects on ABA-based interventions.

In this meta-analysis, we investigated the prevalence and impact of ABA-based interventions for individuals with ASD in Turkey which included a substantial increase in published studies involving diverse targeted behaviors. Despite varying effect size interpretations based on tau and PCES, our study indicated the growing popularity and potential effectiveness of ABA-based interventions in Turkish culture, aligning with previous research (i.e., Sucuoglu & Demir, 2012). Moderator variables, such as age, gender, and the presence of comorbid disability, did not have a significant effect on the outcomes of the intervention. However, parent-implemented interventions and interventions conducted in homes appeared to have a notable moderating effect, warranting further rigorous investigation within the Turkish context.

Even though the current literature seems *promising*, several areas of improvement are noted in research and practice. First, the acceptability of ABA interventions in Turkish culture still needs to be determined. Considering the significance of culture in guiding ABA practices, future studies should consider comparing how Turkish cultural practices influence the effectiveness of ABA interventions to how western cultural practices influence the effectiveness of ABA interventions Additionally, the social validity of ABA-based interventions in Turkey should be assessed to help conceptualize the social importance of behavior change, implemented procedures, and outcomes for society (Wolf, 1978).

Second, the overall quality of ABA-based studies appeared to be weak based on WWC DSs, given that 28 studies did not meet DSs, 41 studies met standards with reservations, and only two studies (i.e., Bozkus-Genc & Yucesoy-Ozkan, 2021; Kiyak & Tekin-Iftar, 2022) met all DSs without reservations. One main issue was the limited amount of data collected. In the event of intervention attrition, WWC suggests collecting five data points for each condition (Kratochwill et al., 2013) to help demonstrate baseline stability and allow for a long-term study of intervention effects on behavioral outcomes (Dixon et al., 2016). Future research should include more data points (i.e., five or more) to demonstrate significant experimental control and rigorous internal validity (Martella et al., 2013) because we discovered an insufficient number of data points in the baseline and intervention conditions for the majority of studies. Additionally, studies did not demonstrate an intervention effect at three different points in time for ABAB designs and five attempts for alternating treatment designs.

Third, despite the improvements in the scope of ABA-based interventions in the last decade, the effects of interventions targeting important skills such as challenging behaviors remain limited. According to recent analyses, the prevalence of challenging behaviors is 94% among individuals with ASD and 93.7% among individuals with comorbid ASD and ID (Simó-Pinatella et al., 2019). Research demonstrates the detrimental effects of challenging behaviors on the lives of individuals with ASD and their families in Turkey, such as interference with the acquisition of academic and social behaviors, exclusion from educational environments, and lower life quality (Güleç-Aslan, 2020; Koydemir-Özden & Tosun, 2010). Unfortunately, the implementation of ABA-based interventions to decrease challenging behaviors remains sparse in Turkey, while interventions such as functional communication training and differential reinforcement procedures (e.g., DRO, DRI, DRA) have been identified as evidence-based practices to decrease challenging behaviors in western countries (David et al., 2022; Gregori et al., 2019; Lory et al., 2020). The majority of included studies ($n = 17$) involved errorless teaching methods to teach new skills, such as riding a bike, playing tennis, and Turkish-art academic content. Future studies should focus on ABA-based interventions that have been defined as evidence based in western countries (e.g., antecedent-based interventions, extinction) to decrease the challenging behaviors and increase socially acceptable replacement behaviors of individuals with ASD in Turkey.

Fourth, the inclusion of adolescents and adults with ASD was sparse in the Turkish literature. ABA-based interventions have been found to be effective in improving the overall quality of life for adolescents and adults with ASD in western countries (e.g., Burke et al., 2010; Edrisinha et al., 2011), although the number of investigations with adults is smaller than those that include primary and elementary school students (David et al., 2022). A meta-analysis conducted by Roth et al. (2014) included a total of 43 studies with 50 adolescents (12 to 18 years old) and 60 adults (18 years or older). The findings of this meta-analysis revealed medium

to strong effect sizes for interventions addressing academic skills, phobic avoidance, and vocational skills. Additionally, interventions for adaptive skills, problem behavior, and social skills had medium effects regardless of the cognitive functioning of the participants (Roth et al., 2014). However, the current review demonstrated that there is litttle to no focus on adolescents ($n = 8$) and adults ($n = 0$) with ASD in Turkey. One possible reason is the lack of transition services and legal mandates for adolescents and adults with disabilities in Turkey (Karal, 2021). Given that the number of adolescents with ASD approaching adulthood continues to rise, evaluating the empirical support for behavioral interventions for adults with ASD is critically important (Gregori et al., 2019) in Turkey. Therefore, future behavior analytic studies involving adolescents and adults are warranted.

Fifth, the majority of interventions were implemented in segregated settings. Given that the inclusion of students with disabilities is poor in quality and quantity in Turkey (Bakkaloglu et al., 2018; Cakiroglu & Melekoglu, 2014), it is not surprising that the majority of interventions were conducted in individualized instructional settings ($n = 28$) or at gyms/swimming pools ($n = 4$) where the interventions were delivered in one-on-one settings. Unfortunately, implementing interventions within these settings does not promote the development of skills and behaviors necessary to increase access to inclusive environments (Plavnick et al., 2015; Sulu et al., 2021). Thus, there is a need for future research to implement ABA-based interventions in inclusive environments (e.g., Kiyak & Tekin-Iftar, 2022) in Turkey.

Sixth, the majority of the implementations were carried out by researchers (some of them were highly trained Ph.D. students) ($n = 35$) without any training for teachers. One of the reasons for this limitation is the lack of resources for teacher education in Turkey (Bakkaloglu et al., 2018; Karal, 2021), and the recent ABA literature seems to be limited in addressing this issue. This limitation is a concern given that research indicates that interventions for individuals with developmental disorders are not as effective in communities as they are in research settings (Weisz et al., 2005) and do not maintain over time (Storch & Crisp, 2004) due in part to a lack of training of educators and parents (Dingfelder & Mandell, 2011; Sulu et al., 2022). Early incorporation of contextual elements, including the planned practitioner, is required by ASD intervention development guidelines (Dingfelder & Mandell, 2011). Additionally, not using evidence-based practices by practitioners was defined as one of the main gaps between research and practice in Turkey (Karal, 2021). Thus, future research should include a careful evaluation of educator implementation to confirm the presumed feasibility of delivering behavior-analytic interventions by teachers and parents with high fidelity (Lory et al., 2020). Additionally, there is a need to train educators and parents to fade intervention components. Therefore, researchers should integrate teachers (e.g., Kiyak & Tekin-Iftar, 2021; Tunc-Paftali & Tekin-Iftar, 2021) and parents (e.g., Besler & Kurt, 2016; Tekin-Iftar, 2008) into the implementation of ABA-based interventions to increase the external validity of these interventions (Sulu et al., 2022).

Finally, none of the studies ($n = 27$) that had mastery criteria provided a description of how the mastery criterion was determined. On the other hand, one of the dimensions of ABA (i.e., technological) requires that researchers describe the procedures explicitly (Baer et al., 1968) to increase the applicability and the replication of the interventions successfully (McDougale et al., 2020). Future studies should operationally define how mastery criteria are set for the expected behavioral outcomes based on each individual's performance level and Turkish culture. Such studies should also collect data until consistent behavioral change occurs (Aydin & Tanious, 2021).

In conclusion, our meta-analysis has revealed significant growth in the number of ABA-based intervention studies for individuals with ASD in Turkey, including a variety of targeted behaviors and interventions. The overall effectiveness of these interventions was supported by the effect size analyses, despite some variations. Further investigation is required to promote rigorous research practices, particularly in study quality, inclusion of adolescents and adults, implementation in inclusive environments, training for educators and parents, and cultural acceptability. Addressing these areas of improvement will contribute to a more comprehensive understanding of the effectiveness of ABA-based interventions for individuals with ASD in Turkey and provide valuable insights for educators and behavior analysts in the region.

Data Availability The corresponding author can provide all original documents upon request.

Declarations

Ethical Approval No human participants were involved in any parts of this study.

Consent to Participate No informed consent was needed to complete the study.

Conflict of Interest The authors declare no competing interests.

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